

The Impact of Marriage Age and Literacy on Family Size*

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February 9, 2025

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1 Introduction

Family size is an important demographic variable influencing the economy, public policy decisions and the society. Understanding the determinants of family size can help the economist and policymakers to make better resource allocation and population sustainability. This study investigates the influence of literacy and age at marriage on family size, using data from the World Fertility Survey conducted in Portugal between 1979 and 1980. These factors were selected due to their significant associations with birth rate, as highlighted in previous research.

*Code and data are available at: https://github.com/HaoweiFan0912/family_size_analysis

Existing literature emphasizes the importance of socio-economic and cultural factors in shaping family size. Ahmed and Roy (2023) found that variables such as family income, educational status, and contraceptive use significantly impact household size in Bangladesh. Their findings emphasize the role of education and socio-economic conditions, which align with this study’s focus on literacy as a determinant of family size. Nitsche et al. (2021) expanded this understanding by analyzing how educational pairings influence fertility in European countries. The research also shows that couples with higher educational attainment often have smaller families. Additionally, Bumpass and Mburugu (1977) highlighted the effects of age at marriage on family size, illustrating that marrying at very young or older ages significantly alters fertility outcomes.

This study examines how literacy and age at marriage jointly affect family size in Portugal, aiming to providing a more detailed understanding of these relationships base on existing researches. To achieve this objective, the study employs a Poisson generalized linear model, with a negative binomial model as an alternative, applying an appropriate offset where necessary. These models enable the modeling of relationships between predictor variables and a discrete, non-negative dependent variable. By using this method, the study quantifies the effects of these predictors on family size, bridging gaps identified in previous research.

2 Methodology

To study family size, assume that each household in the sample has the same number of elder family members for there is no direct variable in the dataset that represents this concept. This assumption allows us to use “Number of Children” as a proxy for family size and treat it as our response variable. The number of children is clearly a positive integer, we first considered using Poisson regression as our modeling approach, which is a type of generalized linear model (GLM) designed for count data. The Poisson regression model includes literacy and age at marriage as predictors. This is because these two variables has been shown significantly influence family size in prior studies in the Section 1. To account for varying exposure times across observations and ensure model fairness, an offset variable is introduced to standardize the response. Specifically, we divide “Number of Children” by this offset, transforming our response variable into a ratio-based measure, ensuring that the model reasonably explains the event occurrence under a per-unit exposure framework.

Poisson regression assumes the mean and variance are equal, which can lead to poor performance when applied to overdispersed data—i.e., when the variance exceeds the mean. To address this, we use a Negative Binomial model, which retains the same link function, offset, and variable levels but assumes the number of children follows a Negative Binomial distribution instead of a Poisson distribution.

Upon validating the models’ applicability, we conducted a hypothesis test to evaluate the statistical significance of the models’ predictions. The hypotheses are formulated as follows:

$$H_0 : \beta_i = 0, \quad H_1 : \beta_i \neq 0$$

where β_i denotes the regression coefficient of the predictor variable under examination.

The statistical inference is based on the z-statistic, computed as:

$$z = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)} \sim N(0, 1)$$

where $\hat{\beta}_i$ represents the maximum likelihood estimate of β_i , and $\text{SE}(\hat{\beta}_i)$ is the standard error of the estimated coefficient.

To quantify the uncertainty associated with the parameter estimates, the 95% confidence interval (CI) for the exponentiated coefficient, which corresponds to the rate ratio in Poisson regression, is given by:

$$\exp(\text{CI}_{95\%}) = \left[\exp\left(\hat{\beta}_i - 1.96 \times \text{SE}(\hat{\beta}_i)\right), \exp\left(\hat{\beta}_i + 1.96 \times \text{SE}(\hat{\beta}_i)\right) \right]$$

If the confidence interval excludes 1, the predictor variable has a statistically significant effect on the response variable at the 95% confidence level in the exponentiated model summary. If the model summary is in log form, statistical significance is determined by whether the confidence interval excludes 0.

3 Result

3.1 Important Variables

In Table 1, the number of children in the sample is right-skewed, with a median and mean of 2. Most families are small, with an interquartile range (IQR) of 1 to 3 children, while a maximum of 17 suggests the presence of extreme outliers. The Figure 1 confirms this skewed distribution, with the majority having few children.

The months since marriage variable (Table 1) has a median of 148 months (~12.3 years) and a wide spread from 0 to 431 months. More recent marriages are more frequent, as shown in the Figure 1. The age at marriage peaks at 22-25 years, indicating this is the most common range, while a slight right-skew reflects later marriages. Literacy is widespread, with 4567 literate individuals compared to only 581 illiterate, as seen in the bar chart (Figure 1).

Table 1: Summary of Key Variables: The number of children has a median of 2 and is right-skewed. Median marriage duration is 148 months, ranging from 0 to 431 months. Most participants married at 22–25 years. 4,567 participants are literate compared to 581 illiterate.

Number of Children	Months Since Marriage	Count by Age at Marriage	Count by Literate or Illiterate
Min: 0	Min: 0.0	At 0-15: 52	Literate: 4567
1st Qu: 1	1st Qu: 73.0	At 15-18: 452	Illiterate: 581
Median: 2	Median: 148.0	At 18-20: 910	
Mean: 2	Mean: 155.2	At 20-22: 1126	
3rd Qu: 3	3rd Qu: 230.0	At 22-25: 1468	
Max: 17	Max: 431.0	At 25-30: 923	
		At over 30: 217	

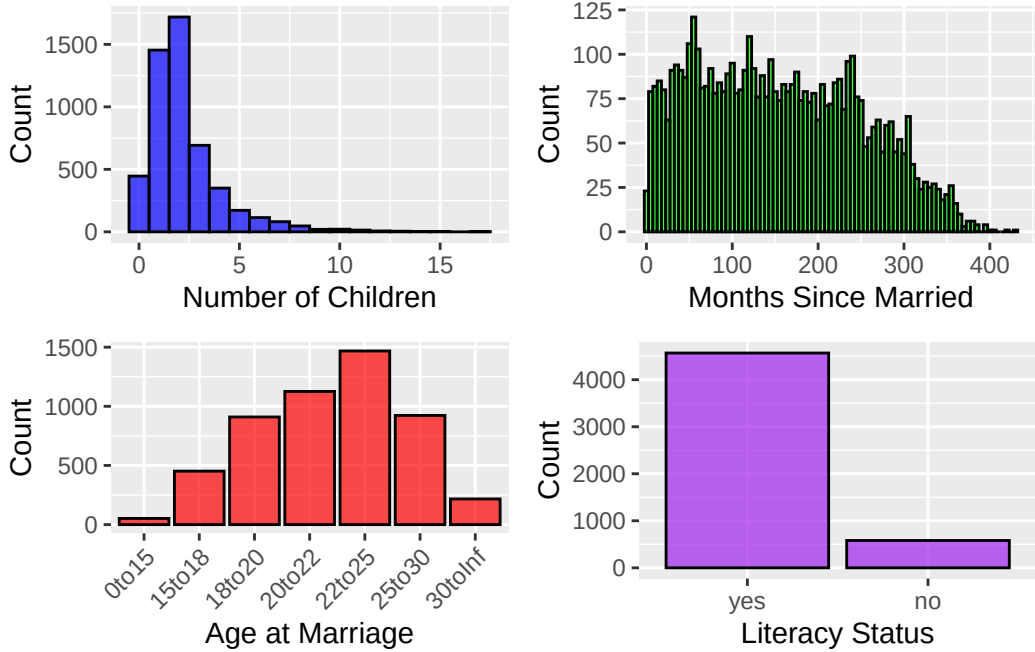


Figure 1: Distributions of Key Variables: Number of Children, Months Since Marriage, Age at Marriage, and Literacy Status: the number of children is right-skewed, marriage duration is relatively evenly distributed within a few hundred months after marriage, the peak age at marriage is 22–25 years, and the majority of individuals are literate.

3.2 Model Selection

Initially, we considered using Poisson regression because the number of children, as the response variable, is a non-negative integer. Predictor variables include “Age at Marriage” and “Literate or Illiterate.” In Todesco’s study, we also account for the significant effect of marriage duration on fertility by using “Months of Marriage” as an offset (Todesco 2011). This ensures the model isolates the effects of literacy and age at marriage by standardizing the exposure period.

Since the Poisson model uses a log link function, taking $\log(0)$ for the offset would result in negative infinity, which would affect the model stability. To address this, we set a minimum of 1 month for marriage duration and converted months to years (dividing by 12) for a more interpretable time scale. The offset transforms the response variable (number of children) into a rate, estimating the fertility rate per year after marriage. With the log link function, results are interpreted as the log of the average number of children per year.

Furthermore, based on Table 1, we observed that the majority of our sample got married between the ages of 22 and 25. Therefore, we set the base level for the “Age at Marriage” variable to the “22 to 25 years” group. This decision allows us to interpret the effects of other age-at-marriage categories relative to the most common marriage age group in our sample.

Thus, our poisson model is:

$$\begin{aligned} \log(E[\frac{\text{Children}_i}{\text{Years}_i}]) = & \beta_0 + \beta_1 M_{0to15}(i) + \beta_2 M_{15to18}(i) + \\ & \beta_3 M_{18to20}(i) + \beta_4 M_{20to22}(i) + \beta_5 M_{25to30}(i) + \\ & \beta_6 M_{30toInf}(i) + \beta_8 \text{Literacy}_i + \\ & \epsilon_i \end{aligned}$$

Where:

$\text{Children}_i \sim \text{Poisson}(E(\text{Children}_i))$

Children_i : The total number of children born to the i th individual.

$M_{atob}(i)$: Binary indicator variable, taking the value of 1 if the i th individual entered into marriage between ages a and b , and 0 otherwise.

Literacy_i : Binary indicator variable, equal to 1 if the i th individual is illiterate, and 0 otherwise.

Year_i : The total duration of marriage (in years) for the i th individual.

$\beta_0, \beta_1, \dots, \beta_8$: Coefficients estimated in the regression model.

ϵ_i : Stochastic error term capturing unobserved heterogeneity and random fluctuations.

Since Poisson regression lacks a variance parameter and assumes that the mean and variance are equal, it may not perform well when applied to overdispersed data. By calculation, the mean number of children is 2.26, while the variance is 3.464, resulting in a variance-to-mean ratio of 1.532. This indicates that the variance is considerably larger than the mean, suggesting the presence of overdispersion. Therefore, to better model data with excess dispersion, we use Negative Binomial Regression as an alternative model:

$$\begin{aligned} \log(E[\frac{\text{Children}_i}{\text{Year}_i}]) = & \beta_0 + \beta_1 M_{0to15}(i) + \beta_2 M_{15to18}(i) + \\ & \beta_3 M_{18to20}(i) + \beta_4 M_{20to22}(i) + \beta_5 M_{25to30}(i) + \\ & \beta_6 M_{30toInf}(i) + \beta_8 \text{Literacy}_i + \\ & \epsilon_i \end{aligned}$$

Where:

$\text{Children}_i \sim \text{Negative Binomial}(E(\text{Children}_i), \theta)$

Others : The rest of the notation is the same as in the Poisson model.

3.3 Model Results

The birth rate ratios from the Poisson model, shown in Figure 2, compare different groups to the reference group (ages 22-25, birth rate ratio = 1.00), with the model accounting for marriage duration by estimating birth rates per unit time. Individuals who married between ages 15-18 (1.079, 95% CI: 1.009-1.153) and 18-20 (1.064, 95% CI: 1.007-1.123) have significantly higher birth rates than the reference group, as their confidence intervals do not include 1.

Individuals married at ages 0-15 (1.051, 95% CI: 0.896 - 1.224) or above 30 (1.022, 95% CI: 0.909 - 1.145) have slightly higher birth rates, but their confidence intervals include 1, indicating that this effect is not statistically significant. For individuals who married between 20-22 and 25-30, the estimated birth rate ratios remain close to 1, and their confidence intervals include 1, meaning no statistically significant difference from the reference group.

For literacy, compared to literate individuals (reference group, birth rate ratio = 1.00), the estimated birth rate for illiterate individuals is 1.173 (95% CI: 1.120 - 1.228, Table 2). Since the confidence interval does not include 1, this suggests a statistically significant association between illiteracy and a higher birth rate.

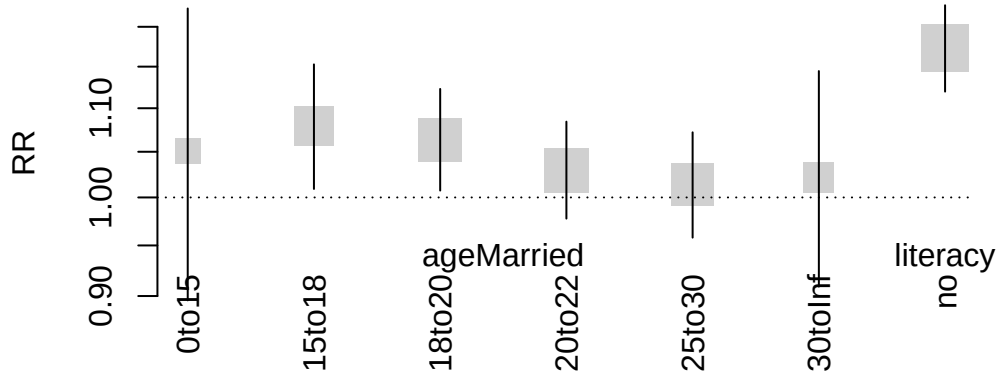


Figure 2: Relative Risk of Number of Children by Age at Marriage and Literacy Status: age groups 15-18 and 18-20 have RRs significantly above 1, and the illiteracy status has an RR significantly above 1, indicating notable differences compared to their reference groups.

Table 2: Summary of The Poisson Model: Marriage between ages 15-18 and 18-20, as well as the illiterate group, have statistically significant positive coefficients and CI do not include 1.

	Estimate	Std. Error	z value	Pr(> z)	est	2.5 %	97.5 %
(Intercept)	-1.802	0.018	-99.076	0.000	0.165	0.159	0.171
ageMarried0to15	0.050	0.080	0.624	0.533	1.051	0.896	1.224
ageMarried15to18	0.076	0.034	2.236	0.025	1.079	1.009	1.153
ageMarried18to20	0.062	0.028	2.228	0.026	1.064	1.007	1.123
ageMarried20to22	0.029	0.026	1.105	0.269	1.030	0.978	1.084
ageMarried25to30	0.013	0.029	0.468	0.640	1.014	0.958	1.072
ageMarried30toInf	0.022	0.059	0.369	0.712	1.022	0.909	1.145
literacyno	0.159	0.024	6.770	0.000	1.173	1.120	1.228

Table 3 summarizes the Negative Binomial model estimating the log rate of the number of children per year. Marriages at ages 15-18 (coefficient = 0.084, 95% CI: 0.009 to 0.158) and 18-

20 (coefficient = 0.069, 95% CI: 0.009 to 0.129) are associated with significantly higher log birth rates, as their confidence intervals exclude 0. In contrast, marriages at ages 0-15, 20-22, 25-30, and above 30 have confidence intervals that include 0, indicating no significant differences from the reference group. Illiterate individuals (coefficient = 0.148, 95% CI: 0.095 to 0.200) show a significantly higher log birth rate compared to literate individuals. The standard deviation parameter (0.265, 95% CI: 0.235 to 0.298) indicates moderate overdispersion, supporting the use of the Negative Binomial model over Poisson regression.

Table 3: Summary of The Negative Binomial Model: Marriage between ages 15-18 and 18-20, as well as the illiterate group, have statistically significant positive coefficients and CI do not include 0.

	Estimate	97.5 %	2.5 %
(Intercept)	-1.783	-1.744	-1.822
ageMarried0to15	0.068	0.244	-0.109
ageMarried15to18	0.084	0.158	0.009
ageMarried18to20	0.069	0.129	0.009
ageMarried20to22	0.035	0.092	-0.022
ageMarried25to30	0.011	0.072	-0.051
ageMarried30toInf	0.021	0.144	-0.102
literacyno	0.148	0.200	0.095
sd	0.265	0.235	0.298

4 Conclusion

The results of this study demonstrate a significant relationship between age at marriage, literacy status, and family size. Both the models reveal a similar result, while the standard deviation parameter in Table 3 indicates moderate overdispersion, we draw the conclusion with the Negative Binomial model. Illiterate individuals have an expected log childbearing rate 0.148 higher than literate individuals (95% CI: 0.095 to 0.200), indicating a statistically significant effect. Therefore, literacy plays a significant role in family size, with illiterate individuals having larger families. This aligns with findings by Ahmed and Roy (2023) and Nitsche et al. (2021), which are introduced in detail in the Section 1. Age at marriage has a mixed effect. individuals who married between 15-18 and 18-20 years old have significantly higher log childbearing rates compared to the reference group (22-25 years old), with coefficients of 0.084 and 0.069, respectively. However, individuals who married at ages 0-15, 20-22, 25-30, or above 30 show no statistically significant differences, as their confidence intervals include 0. These results indicate a positive association between earlier marriage and childbearing rate compared to their corresponding reference group, which aligns with finding by Bumpass and Mburugu (1977).

These findings suggest that early marriage and lower literacy levels are strongly associated with larger family sizes, which aligns with literature's findings in [Section 1](#) that emphasize the influence of education and socio-economic conditions on family size. These results contribute to a deeper understanding of demographic determinants of family size. Economists and policymakers can use these findings to make better decisions regarding resource allocation, such as education and health resources, and to promote population sustainability by managing fertility rates.

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